HC HANDBOOK

#casestudy

Design and interpret case studies

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Defining the HC

Pitch

Case study research designs provide depth of knowledge.

Paragraph Description

A case study focuses on a single example drawn from a population. This sort of study can rely on any of the empirical methods, ranging from observational to experimental. For example, one could study a single planet (e.g., Earth) to learn about properties of planets in general, a single parrot to learn about the limits of avian speech, or a single community to explore how implementing a new homeless policy affects the incidence of homelessness. Case studies are a useful way to refine hypotheses and to gain deeper insights, but care must be taken when generalizing from the results of a case study to a population as a whole.

Categorization

Foundational Concept | Empirical Analyses | Thinking Creatively | Applying Research Methods

Cornerstone Introduction

Class: Empirical Analyses | Semester Two

Unit: Research Design

Big Question: Why do we age?

General Example

You are a researcher investigating human happiness: your goal is to find out what makes some people report more happiness than others. Instead of a variable-led observational study that would collect data from a large sample of individuals to determine whether a variable like income relates to happiness, you decide to use the case study approach to generate hypotheses about human happiness. After you and your research associates conduct extensive literature reviews to understand what factors might influence happiness and the various ways to measure happiness, you embark on data collection. First, you choose Mathieu Ricard, known as "The Happiest Man in the World", as a subject for a case study. You use a range of data collecting techniques, including interviews, behavioral observations, and fMRI brain scans to explore why he might exhibit such a notoriously high level of happiness. For a second case study, you choose a city in Denmark that is known for its happy citizens. You investigate a range of things about the city that might explain the happiness of its residents, such as municipal policies related to quality-of-life and the economic prosperity of the city. Both case studies reveal useful findings, but you are uncertain to what extent the findings can be generalized. You thus decide to test the hypotheses generated by your two case studies using methods from #observationalstudy, such as a prospective cohort study to corroborate how generalizable the findings are.

Footnote: *The researchers choose subjects appropriate for their object of study, happiness. They conduct an extensive literature review prior to choosing variables and data collection methods. They use caution in generalizing from their results, and consider how both #casestudy and #observationalstudy could be used to follow up on their results to see if they are broadly generalizable.*

Is it this HC?

2

Is #casestudy the right HC? If the answer to the following questions is yes, #casestudy could be useful:

1. Are you doing an in-depth case analysis to find insights for your research question?
2. Are you designing a research framework based on a case?

Depending on the focus of the work, there might be other applicable HCs to consider. Consult the following table listing possible related HCs.

	The following HC might apply if the work:
#interventionalstudy	• Designs or interprets an interventional study, usually with large sample sizes. • Attempts to infer a causal relationship among a small number variables. Case studies have small sample sizes.
#observationalstudy	• Designs or interprets an observational study, usually with large sample sizes.
#hypothesisdevelopment	• Develops a hypothesis with sufficient justification from exploratory case studies.
#heuristics	• Uses the “accounting for deviations from the general trend” creative heuristic to generate a solution or a hypothesis.
#studyreplication	• Conducts a partial replication of a case study to investigate the generalizability of the case study’s findings. This step is usually after doing a case study.

2

📝 Self Study

Try answering the questions on your own before checking the example answers.

👆 **When are case studies appropriate?**

Answer: Case studies are used for in-depth investigation of a hypothesis. We utilize case studies when the number of relevant variables is large but the sample size is low. Such conditions typically happen under three scenarios:

- Exploratory case studies (doing a case study before having a hypothesis)
- Complex systems (non-linear interactions among many variables)
- Outliers (they deviate from the general trend, where the majority of data points are)

👆 **What are the differences between a case study and an interventional/observational study?**

Answer: A case study can be either interventional or observational, depending on the context of the research question. The key difference is the sample size. Case studies are distinguished by small sample sizes, whereas typical observational studies have a much larger sample size.

Another difference is the number of dependent variables being explored. A case study looks at many variables, which differentiates it from other studies with small sample size that only consider a few variables. In case studies, such as those for complex systems, multiple significant relationships exist between variables.

2

👆 **What are the differences between a case study and a**

literature review?

Answer: A literature review is an analysis and/or a critique of one or more academic papers. It demonstrates an understanding of existing academic material in a specific field. A case study goes beyond a literature review to explore a set of questions (or in science, likely a set of variables) that needs to be answered in order to evaluate the central question or hypothesis. A quality case study usually starts with a literature review; studying multiple papers can help with the research design, such as identifying relevant variables or questions and constructing data collection methods.

👆 What are the best practices of a case study?

Answer:

1. Start with a literature review on the research question. Consider alternative hypotheses.
2. Choose a case that has a meaningful connection to the research question. Disclose how and why a particular case was chosen.
3. Identify a list of variables to collect or key set of questions to explore.
4. Methods should include how variables or questions will be evaluated.
5. Consider which elements of your case might and might be generalizable and why. This will likely require additional research outside of your particular case.

👆 What are the limitations of a case study?

Answer: The obvious drawback is the low sample size, which makes a case study prone to:

2

- Attentional bias (e.g., paying attention to only certain variables)

- Availability bias (e.g., selecting cases only because they are recent, neglecting older cases which may be better candidates)
- Confirmation bias (e.g., developing the data analysis methods based on the case after collecting data)
- Low statistical generalizability

However, increasing the sample size cannot completely overcome these limitations. We are simply claiming that case studies are more susceptible to these problems.

👆 **Should every case study include a complete research design?**

Answer: Ideally, yes! The terminology "case study" is used in many disciplines from natural sciences to social sciences to business. However, a case study in any discipline would benefit from a systematic methodology, research question, and clear analytical methods from the onset of the study. For example, the French revolution could be a good case candidate if we were to study the causal factors of revolutionary war. Outlining how you will evaluate this research question including use of different sources and *how* the impacts of various factors will be evaluated strengthens the conclusions. A literature review summarizing ideas on what has led to the French revolution is not a #casestudy (but perhaps can be the first step). See more examples below!

Applying the HC

Guided Reflection

- Have you identified a suitable case with a sufficient explanation of its connection to the research question?
- Have you identified relevant variables or key questions, data collection, and data analysis methods?
- Have you explained how the exploratory case study leads to the hypothesis?
- Have you discussed the generalizability of the case study (usually as future work)?

2

Common Pitfalls

- The application does not have the key required aspects of #casestudy: systematic methodology - e.g., which questions or variables are being studied and why?
- The application mentions a specific scenario from a typical interventional/observational study, then assumes it is a case study.
- The application misinterprets a literature review as a case study.
- The application does not show the relevance of the case study to the research question/ hypothesis.
- The application does not discuss the generalizability of the case study.

Applying the HC at Minerva

- In psychology, case studies on rare conditions are often used to explore research questions. An example discussed in Empirical Analyses is a patient's memory loss after a viral infection of the hypothalamus. Such neurodivergent disorders usually have low sample sizes, thus performing a large-scale interventional/observational study is not feasible. These case studies can provide valuable information about brain function that would otherwise be impossible to obtain.
- In Earth Sciences, the Earth itself is often used as a single case to enable generalizations to other planets. As a very simple example: on Earth, life requires liquid water. Thus, the search for extrasolar life has begun with the constraint that life requires water, constraining which planets are candidates for life based on temperature considerations.
- In Philosophy, Ethics, and the Law, case studies are used to help answer more general questions by analyzing a more concrete case in which it applies. Philosophical case studies typically begin with a general research question, like “When should we prioritize public safety over personal freedom?,” and then use a case study to address a more feasible and concrete question within this broad context, like “Obey Law on Criminal Reporting for a Pregnant Mother Addicted to Heroin?”
    - Case study analyses then systematically proceed by first identifying the ethical significance of the case, asking “What makes this case ethically relevant?” For this, 2

students apply normative evidence derived from ethical theories to argue for the relevance of that theory to a particular case.
    - They then typically present multiple possible decisions someone could make, and how each could be ethically justified, considering counter-arguments to those decisions.
    - Finally, case study analyses conclude with the student's defense of their judgment for the best resolution of the dilemma presented by the case study and explain how their response relates to the broader philosophical question they began with.

Applying the HC Outside of Minerva

- Husnayain et al. (2020) investigated the role of artificial intelligence in risk communication during an infectious disease outbreak. They used the COVID-19 outbreak in Taiwan (a complex system) as a case study and used Google Trends to monitor search trends, which reflect the public's awareness of personal risk management. For example, the study found that searches related to face masks increased rapidly following the announcements of the first imported case and peaked when local cases were reported. Meanwhile, searches for handwashing gradually increased in response to the face mask shortage in Taiwan. The study concluded that these search trend dynamics may be generalized to future outbreaks to define the proper timing and location for risk communications (e.g., at what outbreak stage it's appropriate to raise awareness on face masks), leading to more efficient resource allocations.

Key Resources

- Donnelly, Katie. (2022, February 8). *Defining Case Studies as a Research Method*. Minerva University. https://my.minerva.edu/academics/hc-resources/cornerstone-custom-hc-guides/
    - This whitepaper describes when to use case studies, the best practices to design case studies, and a few examples.

2

- Calvet, G., Aguiar, R., Melo, A., Sampaio, S., Filippis, I.D., Fabri, A., ... Filippis, A.M. (2016). Detection and sequencing of Zika virus from amniotic fluid of fetuses with microcephaly in Brazil: a case study. Lancet Infectious Disease, 16: 653–660. (1 semester fair use) https://course-resources.minerva.edu/uploaded_files/mu/00255606-9396/calvetetal-2016-zikavirus-casestudy-annotated--1-.pdf

- This annotated paper outlines a case study on the transmission of the Zika virus from the mother to the fetus, leading to microcephaly. The study was done only on two pregnant women in Brazil.

2